CHAPTER 5

Conditional probabilities and independence

Chapter path

Objectives. Change the reference sample space; build a tree; use a partition; distinguish incompatibility from independence; handle up to four Bernoulli trials.
Prerequisites. Fractions, proportions, probability on a finite sample space, intersection and union.
Yearly reference point. 17-26 November: 8 hours

Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Out of 40 students, 12 practise a given sport. Proportion?

2. If $P(A) = 0{,}3$, give $P(\overline{A})$.

3. Write "A and B" in set notation.

4. Express "A or B" with the inclusive or.

5. Calculate $0{,}4 \times 0{,}25$.

6. A fair die is rolled. Give $P(\{2; 4; 6\})$.

After the diagnostic check. Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

Oral work and logic. Explain aloud what "given", "and" and "at least" mean. On a tree, justify an addition and a multiplication.

Investigate before formalising

Build a table of 100 people in which A and B are independent, with 40 A and 30 B.
Change two cells while keeping these margins so as to break independence.

Organisation. Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min.
Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

5.1 Conditioning: changing the denominator

Definition and results

For $P(A) > 0$, the probability of B given A is $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. In a table of equally likely individuals, it is the ratio of the count of $A \cap B$ to the count of A. In general $P_A(B) \neq P_B(A)$.

Status and justification. Definition. Multiplying by $P(A)$, we get the rule $P(A \cap B) = P(A)P_A(B)$. The condition $P(A) > 0$ is essential.

Worked example. Out of 100 people, 40 belong to A, 30 to B and 12 to both. $P_A(B) = \dfrac{12}{40} =$ 0,3, while $P_B(A) = \dfrac{12}{30} = 0{,}4$. $P(A \cap B) = 0{,}12$.

Watch out

"Among the A" sets the denominator. A conditional probability is not an intersection.

Immediate application. Out of 60 objects, 20 are red; of the red ones, 5 are damaged. Calculate $P(R \cap D)$ and $P_R(D)$.

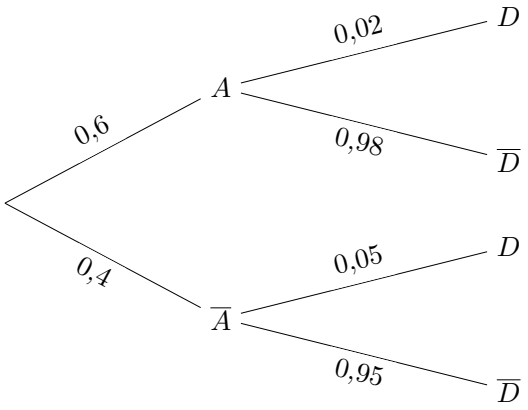
5.2 Trees, partitions and total probability

Definition and results

A partition $A_1, \ldots, A_k$ is a family of disjoint events whose union is the sample space. If their probabilities are positive, $P(B) = \sum_i P(A_i)P_{A_i}(B)$. On a tree, the probabilities on branches from the same node sum to 1; probabilities along a path are multiplied; probabilities of disjoint paths are added.

Status and justification. B is the disjoint union of the $B \cap A_i$. By additivity and by the definition of conditional probabilities, $P(B) = \sum_i P(B \cap A_i) = \sum_i P(A_i)P_{A_i}(B)$.

Worked example. Two machines supply 60% and 40% of the output; their defect rates are 2% and 5%. $P(D) = 0{,}6 \times 0{,}02 + 0{,}4 \times 0{,}05 = 0{,}032$. Among defective items, the first machine represents $\dfrac{0{,}012}{0{,}032} = \dfrac{3}{8}$.



A denotes the machine producing 60 per cent of the output and D a defect. For example, $0{,}02 = P_A(D)$: the second-level probabilities are conditional.

Watch out

The simple mean of the rates is correct only if the branch weights are equal.

Immediate application. $P(A) = \frac{1}{3}$, $P_A(B) = \frac{3}{4}$, $P_{\overline{A}}(B) = \frac{1}{4}$. Calculate $P(B)$.

5.3 Independence and disjointness

Definition and results

A and B are independent if $P(A \cap B) = P(A)P(B)$. If $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$. They are mutually exclusive if $A \cap B = \varnothing$. Two mutually exclusive events with strictly positive probabilities are not independent.

Status and justification. The conditional criterion follows from dividing by $P(A) > 0$. For mutually exclusive events with positive probabilities, the left-hand side is 0 and the product on the right is strictly positive.

Worked example. With a fair die, A: even result, B: result at least 5. $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{6}$: independent. A and "odd result" are mutually exclusive but not independent.

Watch out

Independence in a model must be justified or stated explicitly; "two different events" is not enough.

Immediate application. We are given $P(A) = 0{,}4$, $P(B) = 0{,}5$, $P(A \cap B) = 0{,}18$. Are they independent?

5.4 Successive trials and Bernoulli

Definition and results

A Bernoulli trial has two outcomes: success, with probability p, and failure, with probability 1 $-p$. In a repetition of n identical independent trials, with $n \leq 4$ here, the probabilities of the branches remain p and 1 $-\,p$. A path with k successes has probability $p^k(1-p)^{n-k}$; we counts the paths that realise the event, without the general theory of the binomial distribution.

Status and justification. The probability of a path is a product, by independence. The paths corresponding to an event are disjoint: their probabilities add up.

Worked example. For three trials with $p = \frac{1}{2}$, exactly two successes correspond to SSE, SES, ESS. Each path has probability $\frac{1}{8}$, so the total is $\frac{3}{8}$. At least one success has probability $1 - (\frac{1}{2})^3 = \frac{7}{8}$.

Watch out

When drawing from an urn without replacement, the probabilities generally change after the first draw: the independence model may be wrong.

Immediate application. Four independent trials with success probability $p = \frac{1}{3}$. Calculate the probability of four failures, then of at least one success.

5.5 Simulating and discussing the model

Definition and results

A pseudo-random simulation produces frequencies that fluctuate around a probability of the model. For a uniformly distributed point in the square [0; 1] ^{2}, the proportion satisfying $y \leq x^2$ estimates the area under the parabola; the proportion satisfying $x^2 + y^2 \leq 1$, multiplied by 4, estimates π. The repetitions must use new draws.

Status and justification. Simulation principle assumed and tested experimentally. The probability-area equality is an interpretation of the uniform draw; integration is not a prerequisite. A simulation does not prove an exact value.

Worked example. Out of 10000 points, if 7840 are in the quarter disc, the estimate of π is $\frac{4 \times 7840}{10000} = 3{,}136$. Two samples of the same size generally give different values.

Watch out

Fixing a seed makes it possible to reproduce a run, not to remove the randomness of the model or to guarantee a given precision.

Immediate application. A simulation of 2000 points gives 675 points under $y = x^2$. Which area does it estimate?

Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 5.1 · A table

A survey covers 100 users of two transport services A and B. Of these, 40 use A, 35 use B and 15 use both. One of the 100 users is chosen at random; all have the same probability of being chosen. The events A and B respectively mean "the user uses the service A" and "the user uses service B".

1. Construct and complete a two-way table: use of A or not in rows, use of B or not in columns. Add the totals.
2. Calculate $P_A(B)$ and $P_B(A)$. For each calculation, state the reference population used in the denominator.
3. Interpret both probabilities with two separate sentences.

Exercise 5.2 · Production

A company manufactures objects in two workshops A and B, which produce respectively $\frac{1}{4}$ and $\frac{3}{4}$ of the production. Among the items produced by A, 4 per cent are defective; among those produced by B, 2 per cent are. An item is chosen at random from the whole production. Let D the event "the item is defective".

1. Represent the situation with a weighted tree and state the meaning of each probability on the second level.
2. Calculate $P(D)$ and interpret the result as an overall defect rate.
3. We now know that the chosen object is defective. What is the probability that it was made in workshop A?

Exercise 5.3 · Independence to be verified

A fair die with six faces numbered from 1 to 6 is rolled. Consider the events A : "getting an even number" and B : "getting a number greater than or equal to 4".

1. Write out the outcome sets corresponding to A, B and $A \cap B$.
2. Determine whether the events A and B are independent. Justify using a numerical criterion.
3. Explain why they are not disjoint either.

Exercise 5.4 · Without replacement

An urn contains three red balls and two blue balls, indistinguishable to the touch. A first ball is drawn at random, then a second without replacing the first. At each draw, all the balls in the urn are equally likely.

1. Construct the weighted tree of the two draws, distinguishing the colour obtained on the first and on the second draw.
2. Calculate the probability of getting two red balls.
3. Calculate the probability of getting one ball of each colour. Explain why two paths of the tree must be taken into account.

Exercise 5.5 · Three trials

An experiment has a success with probability $\frac{1}{3}$ and a failure. It is repeated three times independently, under the same conditions.

1. Construct a three-level tree, or list the eight possible sequences of outcomes.
2. Calculate the probability of exactly one success. State the paths used.
3. Calculate the probability of getting at least two successes, distinguishing the cases of two and three successes.

Exercise 5.6 · Detector

A device checks objects of which 2 per cent are defective. It triggers a signal for 90 per cent of defective items and, wrongly, for 5 per cent of non-defective items. An item is chosen at random from those checked. Let D the event "the item is defective" and S the event "the device triggers a signal".

1. Determine the proportion of actually defective items among the flagged items. Choose a representation and explain the reasoning in detail.
2. Give an exact value, then a percentage rounded to one decimal place.
3. A person immediately gives 90 per cent as the answer. Explain precisely which confusion between probabilities they are making.

Exercise 5.7 · Sets and cardinality

We work in the set $E = \{1; 2; 3; 4\}$. Consider its subsets $A = \{1; 2\}$ and $B = \{2; 4\}$.

1. Write $A \cap B$, $A \cup B$ and the complement of A in E. Give the cardinality of each.
2. Write all pairs of $A \times B$, then give its cardinality.
3. Explain why the pairs (1; 2) and (2; 1) cannot be confused. Does the second belong to $A \times B$?

Exercise 5.8 · Delayed review

In a random experiment, two events A and B satisfy

$$P(A) = \frac{1}{2}, \qquad P_A(B) = \frac{1}{4}, \qquad P_{\overline{A}}(B) = \frac{3}{4}.$$

1. Calculate $P(B)$ using a tree or the law of total probability.

2. Calculate $P_B(A)$ by showing the common event used in the numerator.

3. The probabilities $P_A(B)$ and $P_B(A)$ equal here? Is this equality a general property ?

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 67 p. 336, questions 1, 2 and 3. Build a tree keeping the events D and N from the textbook. Note: N denotes a negative test.

CQFD, exercise 70 p. 336, questions a and b. Count the four equally likely outcomes of two rolls to refute the proposed equalities.

MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $P_A(B)$ has denominator
   - A. $P(B)$
   - B. $P(A)$
   - C. $P(A \cap B)$
   - D. $1 - P(A)$
2. $P(A) = 0{,}4$, $P_A(B) = 0{,}2$: intersection
   - A. $0{,}08$
   - B. $0{,}6$
   - C. $0{,}5$
   - D. $0{,}2$
3. A partition consists of events
   - A. equal
   - B. all independent
   - C. with the same probability
   - D. disjoint and covering the sample space
4. Independence is equivalent to
   - A. $P(A) = P(B)$
   - B. $P(A \cap B) = 0$
   - C. $P(A \cap B) = P(A)P(B)$
   - D. $A = B$
5. Two disjoint events with positive probabilities are
   - A. independent
   - B. not independent
   - C. equal
   - D. necessarily complementary

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. Three independent tosses of a fair coin, at least one tail:
   - A. $\frac{1}{8}$
   - B. $\frac{3}{8}$
   - C. $\frac{7}{8}$
   - D. $\frac{1}{2}$
2. Two independent trials with $p = \frac{1}{3}$, two successes:

A. $\frac{1}{9}$

B. $\frac{2}{3}$

C. $\frac{1}{3}$

D. $\frac{4}{9}$

3. The branches leaving the same node sum to

A. 0

B. 2

C. the probability of the path

D. 1

4. A simulation gives a frequency of 0.51. This

A. proves $p = 0{,}51$

B. is a fluctuating observation

C. proves $p = \frac{1}{2}$

D. refutes $p = \frac{1}{2}$

5. Drawing without replacement, after a red among 3 red and 2 blue, the probability of red is:

A. $\frac{3}{5}$

B. $\frac{3}{4}$

C. $\frac{1}{2}$

D. $\frac{2}{5}$

True or false? Justify.

1. $P_A(B) = P_B(A)$ for all events with positive probabilities.

2. If A and B are independent, then A and $\overline{B}$ are too.

3. Two successive draws are always independent.

Personal review

I find the right denominator; I check the sums at the nodes; I justify independence; I count the paths without confusing them.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.

Solutions and teaching notes, chapter 5

Teacher use. The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. $\frac{12}{40} = \frac{3}{10} = 30\,\%$. *Action:* Link the partial count to the reference count.
2. $0{,}7$. *Action:* Represent the sample space split into two events.
3. $A \cap B$. *Action:* Shade the intersection of two sets.
4. $A \cup B$. *Action:* Include the common outcomes in the union.
5. $0{,}1$. *Action:* Replace the decimals with fractions.
6. $\frac{3}{6} = \frac{1}{2}$. *Action:* Identify the equally-likely assumption.

Investigation

Intersection needed 12; cells 12,28,18,42. Replacing with 13,27,17,43 keeps the margins but $\frac{13}{100} \neq 0{,}4 \times 0{,}3$.

Direct applications

1. $\frac{5}{60} = \frac{1}{12}$ and $\frac{5}{20} = \frac{1}{4}$.
2. $\frac{1}{4} + \frac{1}{6} = \frac{5}{12}$.
3. No: $0{,}4 \times 0{,}5 = 0{,}20 \neq 0{,}18$.
4. $(\frac{2}{3})^4 = \frac{16}{81}$ then $1 - \frac{16}{81} = \frac{65}{81}$.
5. $\frac{675}{2000} = 0{,}3375$; this is an estimate, not the exact value.

Exercises: methods and answers

Exercise 5.1 - A table. Cells: 15 (A and B), 25 (A without B), 20 (B without A), 40 (neither A nor B). $P_A(B) = \frac{15}{40} = \frac{3}{8}$; $P_B(A) = \frac{15}{35} = \frac{3}{7}$.

Exercise 5.2 - Production. $P(D) = 0{,}25 \times 0{,}04 + 0{,}75 \times 0{,}02 = 0{,}025$. $P_D(A) = \frac{0{,}01}{0{,}025} = 0{,}4$. The denominator changes when conditioning on D.

Exercise 5.3 - Independence to be checked. $P(A) = P(B) = \frac{1}{2}$; intersection $\{4; 6\}$ with probability $\frac{1}{} \neq \frac{1}{}$: not independent. The fact that another choice of threshold gave independence does not automatically carry over.

Exercise 5.4 - Without replacement. Two reds: $(\frac{3}{5})(\frac{2}{4}) = \frac{3}{10}$. One of each: $(\frac{3}{5})(\frac{2}{4}) + (\frac{2}{5})(\frac{3}{4}) = \frac{3}{5}$. The order distinguishes two disjoint paths.

Exercise 5.5 - Three trials. One success: three paths, each $(\frac{1}{3})(\frac{2}{3})^2 = \frac{4}{27}$, total $\frac{4}{9}$. Two success: $3(\frac{1}{3})^2(\frac{2}{3}) = \frac{2}{9}$; three: $1\overline{27}$; total $7\overline{27}$.

Exercise 5.6 - Detector. $P(S) = 0{,}02(0{,}9) + 0{,}98(0{,}05) = 0{,}067$. $P_S(D) = \frac{0{,}018}{0{,}067} = \frac{18}{67} \approx 26{,}9\,\%$. The low initial rate explains the gap from 90%.

Exercise 5.7 - Sets and cardinality. $A \cap B = \{2\}$ (1) ; $A \cup B = \{1;2;4\}$ (3) ; $E \setminus A = \{3;4\}$ (2); product $\{(1;2),(1;4),(2;2),(2;4)\}$ (4). A pair is ordered.

Exercise 5.8 - Delayed review. $P(B) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$. $P_B(A) = \frac{\frac{1}{8}}{\frac{1}{2}} = \frac{1}{4}$. The numerical equality with $P_A(B)$ is specific to these data.

CQFD, exercise 67 p. 336, questions 1, 2 and 3. $P(\overline{D} \cap \overline{N}) = 0{,}9 \times 0{,}05 = 0{,}045$. The probability of a positive test is 0 ,$1 \times 0{,}97 + 0{,}045 = 0{,}142$. Among positive tests, the probability of not being doped is $\frac{0{,}045}{0{,}142} = \frac{45}{142} \approx 31{,}7\%$. Do not swap positive and negative tests.

CQFD, exercise 70 p. 336, questions a and b. $P(B) = \frac{1}{4}$, $P_A(B) = \frac{1}{2}$ and $P_{\overline{A}}(B)$ = 0. Thus $\frac{1}{4} \neq \frac{1}{2}$ + 0, then 0 $\neq 1 - \frac{1}{2}$. The law of total probability involves the weights $P(A)$ and $P(\overline{A})$; the complement applies to B when the same condition is kept.

MCQ: answers and reasons

MCQ 1.

1. B. Reference sample space A.

2. A. Product along the branches.

3. D. Definition.

4. C. Criterion without division.

5. B. Empty intersection, positive product.

MCQ 2.

1. C. Complement of three faces.

2. A. Product p^2.

3. D. Probabilities of a conditional partition.

4. B. Observed frequency and model probability differ.

5. C. 2 red balls remain out of 4.

True or false

1. False: 12 common objects out of 40 A and 30 B give $\frac{3}{10}$ and $\frac{2}{5}$.

2. True: $P(A \cap \overline{B}) = P(A) - P(A \cap B) = P(A)(1 - P(B))$.

3. False: without replacement, the contents of the urn change.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.